

Zongze WU



- Year 3 Ph.D. Student, Sport Science
 - Faculdade de Desporto, Universidade de Porto
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Education

- ✧ Master of philosophy
Physical Education and Sport Studies, University of Macau, Macau, China Supervisor:
Prof. Zhaowei Kong, associate Dean of the Faculty of Education
- ✧ Bachelor of education
Sport Science, Harbin Sport University, Harbin, China
Weighted score: 92.05 (ranking 2nd among all students in the department)

Honors and awards

- ✧ Postgraduate
Excellent Graduation Honors, 2023
- ✧ Undergraduate
National Scholarship (six times, top 3%), 2014 - 2018
Excellent Trainee (Research Division of China Heilongjiang Snow Sports Training Center), 2018

Research skills

- ✧ Portable cardio-pulmonary system (cortex & cosmed), Cardio-pulmonary exercise system (Vmax)
Centrifuge, Hypoxico altitude training systems, Xsens
- ✧ Data analysis: Spss, G*Power

Publications

1. Physiological and Biomechanical Characteristics of Inline Speed Skating: A Systematic Scoping Review. [Zongze Wu](#); Filipa Cardoso; David B. Pyne; Márcio Fagundes Goethel; Ricardo J. Fernandes. DOI: 10.3390/app15147994.
2. Acute Ketone Monoester Supplementation in Young Adults: Modulating Metabolic and Neurocognitive Functions Across Body Weights. Qian Yu; Ka Kit Wong; On Kei Lei; Paulo Armada-da-Silva; [Zongze Wu](#); Jinlei Nie; Qingde Shi; Zhaowei Kong. DOI: 10.1139/apnm-2024 0229.
3. Cerebral oxygenation and cardiac output responses during short repeated-sprints exercise and modulatory effect of glucose ingestion. Paulo A. S. Armada-da-Silva; Hu Mingzhu; [Wu Zongze](#); Wen Linjian; Feng Ruisen; Xinglin Zeng; Zhen Yuan; Zhaowei Kong. DOI: 10.1101/2022.12.05.519099.

Services

- ✧ Postgraduate
Postgraduate house tutor, 2020 - 2023
Teaching Assistant of the PE course, 2020 - 2023
- ✧ Undergraduate
President of the student union, 2016 - 2017

Hobbies

Ski, Baduanjin Qigong, Diabolo, Archery, squash, Skate, Flute

Research projects

✧ Postgraduate

Research Assistant at the Kinesiology Laboratory, 2021 - 2023

- i) Preliminary experiment of the effect of ketone ester supplementation diet on brain plasticity in overweight/obese adults: a neurophysiological and resting functional magnetic resonance study. (2021, Publication no. 3)

Contribution: In charge of testing the maximal oxygen consumption, blood acid, blood glucose, 6s-24s training test, and collecting physioflow data.

Results: Our findings suggest that short sprint exercise increases O₂Hb, HHb, and tHb levels during exertion in parallel with cardiac output. However, in addition to the transient increase in cerebral hemoglobin, a progressive decline in cerebral oxygen saturation occurs during repeated sprints. Glucose ingestion does not alter cerebral hemoglobin responses to sprint exercise but appears to be associated with faster recovery of O₂Hb.

- ii) The effect of High-Intensity Interval Training and moderate-intensity continuous training with a ketogenic diet on physical and psychological changes in obese adults. (2021 - 2023, Publication in preparation)

Contribution: In charge of recruiting subjects, arranging test time, part of the high-intensity interval training and moderate-intensity continuous training test, and collecting pre- and post-VO₂max data and Stroop test data.

Results: Our findings indicate that training with a ketogenic diet had a positive impact on weight loss in overweight and obese young adults. The weight loss from moderate-intensity continuous training with a low-carbohydrate diet was higher than that of the low-carbohydrate diet without exercise and the low-carbohydrate diet with high-intensity interval training groups. However, muscle loss was the largest in the ketogenic diet with the moderate-intensity continuous training group.

- iii) The effect of ketone ester supplementation diet on brain plasticity in overweight/obese adults: a neurophysiological and resting functional magnetic resonance imaging study. (2022 – 2023, Publication no. 2)

Contribution: In charge of subjects' recruitment, blood glucose, blood ketone, and part of the Transcranial magnetic stimulation test.

✧ Undergraduate

- i) Tester of the national ice hockey team cross-major-selection for the 2022 Beijing Winter Olympic Games (The only undergraduate student in the test group) (Harbin, Jilin, Shenyang City), 2018
- ii) Team inspector of the national student fitness standard test (Ji'nan City, Shandong Province), 2017
- iii) Sit in on the international seminar on the winter sports of kinematics (Harbin City), 2016
- iv) Team inspector of the national student fitness standard test (Hohhot City, Inner Mongolia Province), 2015
- v) Member of the university summer practice (Heilongjiang province sports injury rehabilitation hospital), 2015
- vi) Tester of the middle-aged and the elderly physical test (Harbin City), 2014